

Drones Meeting

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

Two delivery drones are flying on a **straight road** in the positive direction.

The first drone starts at position x_1 and moves v_1 meters **every second**.

The second drone starts at position x_2 and moves v_2 meters **every second**.

Both drones start moving at the same time.

Determine whether there exists a positive integer number of seconds after which both drones will be at the same position.

Input

The first line contains four integers x_1 , v_1 , x_2 , and v_2 ($0 \leq x_1 < x_2 \leq 10000$, $1 \leq v_1, v_2 \leq 10000$).

Output

Print YES if the two drones can meet after a positive integer number of seconds. Otherwise, print NO.

Examples

standard input	standard output
0 3 4 2	YES
0 2 5 3	NO